# Does the Benchmark Measure the Physics? An Instrument for Auditing Counterfactual Video Evaluation

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## Abstract

A world model is useful for physical AI only if it gets the physics right, so the benchmarks we use to verify one must reward physical accuracy. We built an instrument to test whether the most widely used counterfactual video benchmark does. It answers CLEVRER's counterfactual questions by explicit intervention, holds every stage fixed except the simulator, and reports the event behind each answer. Swapping four simulators and nine checkpoints across a 2.2× range of rollout error moves the score by 2.2 points; straight-line motion lands within 0.75 points of a learned world model. Two choices containing no physics matter more: how long the world runs past the last frame (4.6 points) and when objects are handed to the simulator (1.2 points). Giving the engine a perfect event detector on the half of the questions it never simulates buys 1.4 points, so sensing is not the hidden explanation either. The flatness is not an artefact of a lenient detector: under a strict contact rule the simulators separate *less*, not more. Physics as a whole is not optional, since removing simulation costs 9.0 points, but accuracy above a low bar buys almost nothing. We argue that a benchmark meant to verify a world model should report, next to accuracy, the share of decisions the simulator actually made and the share whose evidence is valid, and we show both are cheap to measure.

## 1 The question this measures

Physical world AI needs benchmarks that separate a model which predicts the world from one that produces plausible-looking output. That need presumes our benchmarks already reward physical accuracy. We tested the presumption on CLEVRER Yi et al. [2020], the most used benchmark for counterfactual physical reasoning, and found its counterfactual score is close to blind to the accuracy of the simulator producing the answers.

This paper is not a system entered into a leaderboard. It is an instrument built so that one component can be varied while everything else is held fixed, and an audit run with it. The result should generalise past one dataset. Counterfactual and predictive questions compress a continuous trajectory into a discrete outcome: did these two objects touch? That outcome flips only when trajectory error carries a pair across the contact distance; everywhere else, halving the error changes nothing that is scored. Any evaluation graded on outcomes, including manipulation benchmarks scored by task success, shares this structure.

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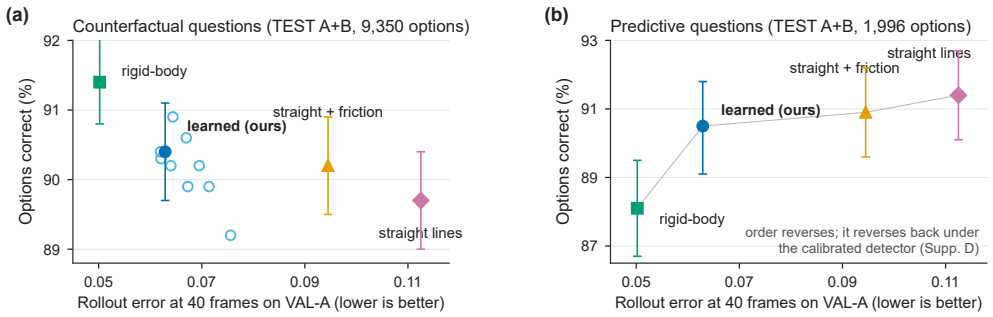



Figure 1: Physics quality against benchmark score. (a) Counterfactual questions: a 2.2× spread in rollout error moves the score by 2.2 points, and most differences sit inside the 95% intervals. Hollow circles are nine further checkpoints of the learned model, so the thirteen points are not independent simulators. (b) Predictive questions: the ordering reverses under the detector chosen on validation and reverses back under a calibrated one, so the ranking tracks the detector rather than the physics. Error on 200 validation videos, scores on 1,398 held-out test videos.

## 2 The instrument

To attribute a score to the simulator, everything else must be held fixed. The engine answers a counterfactual with the three steps of an intervention Pearl [2009]: the recording plays the role of abduction, the edit is the action, and prediction is a simulation. Every object follows its recorded path until the intervention can reach it, is then handed to a simulator a few frames before it would lose a recorded contact, and the world runs on past the end of the video. The engine returns the event that decided each answer: the two objects, the frame, and whether that event was recorded or simulated. It is never trained on questions or answers. Simulating only the objects an intervention affects follows CRCG Ishay et al. [2024]; building one reusable editable world per video is also PhysMind's design Yang et al. [2026].

The simulator is a module. We plug in a learned world model with a pairwise potential, a symplectic integrator and learned per-object friction; an analytic rigid-body simulator whose six constants are fitted on training videos, in the spirit of system identification Ding et al. [2021]; straight-line motion with and without fitted friction; nine earlier checkpoints of the learned model; and six retrained seeds. Rollout error is measured separately, on held-out validation videos, as mean position error at 40 frames.

## 3 What the benchmark rewards

Fig. 1a plots the counterfactual score against rollout error. The error spans 0.050 to 0.113, a factor of 2.2, while the score spans 89.2% to 91.4% of options. The two move in the expected direction ($r = -0.57$, 95% CI $[-0.85, -0.03]$), so the question is not the sign but the size. Nine of the thirteen points are checkpoints of one model, so we read the fitted slope rather than the correlation: 19.4 points of score per unit of rollout error, which over the observed range is 1.2 points. Straight-line motion carries 1.8 times the rollout error of the learned model and scores 0.75 points lower.

The reason is visible in the answers. The four distinct simulators return the same answer on 91.4% of the 9,350 counterfactual options, although at least one asked object is simulated before the video ends on 47.8% of them. Every difference between simulators comes from the remaining 8.6%, which suggests most asked pairs sit far from the contact boundary where a trajectory error could flip an outcome.

This is not an argument that physics is unnecessary. Removing simulation entirely costs 9.0 points per option and 22.2 per question. The claim is narrower: *above a low bar, accuracy buys almost nothing on this metric*.

**What does move the score.** Two choices containing no physics matter more than the choice of simulator. Stopping the simulation at the last recorded frame costs 4.6 points per option and 14.3 per question, because CLEVRER's counterfactual answers include events after the video ends. Handing

Table 1: Counterfactual accuracy on 1,398 held-out videos (9,350 options, 2,619 questions) against rollout error measured separately. Sorted by rollout error.

| Simulator | Rollout error @40 | Options (%) | Questions (%) |
|---|---|---|---|
| analytic rigid-body, constants fitted on training videos | 0.050 | **91.4** | **75.0** |
| learned world model | 0.063 | 90.4 | 72.3 |
| straight lines + friction | 0.095 | 90.2 | 71.9 |
| straight lines | 0.113 | 89.7 | 70.2 |
| no simulation at all | – | 81.4 | 50.1 |
| a 7B vision-language model, 16 frames | – | 54.4 | 2.9 |
| answer "no" to every event | – | 54.1 | 0.0 |

objects to the simulator 12 frames before a lost contact instead of 3 costs a further 1.2 and 3.2 points: every extra frame of real recorded path is drift avoided. A system can gain more from when it starts simulating than from how well it simulates.

# 4 Two explanations we can rule out

**It is not the event detector being lenient.** A natural objection is that our contact rule is generous enough to absorb trajectory error, so the simulators agree because the sensor is blunt rather than because the benchmark is insensitive. That predicts a stricter rule should separate them further. It does the opposite. Re-scoring all four simulators at twelve common hand-off and extension settings, the median spread between them is 2.18 points under the calibrated rule and **1.50 points under a strict one**. Tightening the detector makes the simulators *harder* to tell apart, because every one of them then loses the same real contacts.

**It is not sensing either.** On 52.2% of options neither asked object is ever handed to the simulator, so the counterfactual world is identical to the recording there and a perfect detector is exactly CLEVRER's collision annotation, which the engine never reads. Substituting it raises the learned model from 90.4% to 91.8% and the analytic rigid-body simulator from 91.4% to 92.9% ($z = 9.8$ and 10.2). Perfect detection on half the benchmark buys about what the entire 2.2× sweep of physical accuracy buys. Run blind to everything after the last frame, the same oracle scores 88.2%, *below* the unmodified engine: a perfect detector that cannot see past the video loses to an imperfect one that simulates past it.

Table 1 places the spread in context. The gap between the best and worst real simulator is 1.7 points; the gap between having a simulator and not having one is 9.0; and a general vision-language model given the frames lands at the level of answering "no" to everything.

# 5 Two numbers a benchmark should report

Because the engine returns the event behind each answer, we can ask what the accuracy rests on. Of 8,454 correct answers, 4,973 (58.8%) have a deciding event that can be checked against CLEVRER's annotation, and 98.2% of those agree with it. That number needs its composition stated: roughly half the passing checks confirm that a collision did *not* happen, which is easy in a sparse scene. Counted on positive events only, the detector reaches 93.1% precision at 98.5% recall. The other half of the story is that 40.5% of correct answers are decided in the simulated part of the world, where the dataset has no ground truth at all, so they can be inspected but never verified.

We propose that benchmarks intended to verify world models report two quantities alongside accuracy:

- **the share of decisions the simulator made**, separating answers the recording already settles from answers the physics produced; and
- **the share of correct answers whose evidence is valid**, checked against annotation where it exists, with positive and negative confirmations reported separately.

Both are cheap once a system reports its deciding event, and together they distinguish a model that is right for the right reason from one inheriting its score from the evaluation protocol. On CLEVRER

they reveal that a large, independently verified improvement in physics, a 25% reduction in rollout error from adding dissipation, moves the benchmark by nothing measurable.

**Running this on your own benchmark.** The recipe needs no new annotation. Make the simulator a swappable component and push at least one deliberately weak baseline through the same harness, for instance constant-velocity extrapolation or a rigid-body simulator left at default parameters. If the scored outcome barely separates them, the benchmark is measuring something other than the dynamics, and the separation it does show is how much of the headline number the physics earned. Then, for every scored episode, log which predicted event the decision rested on and whether that event occurred in the real rollout. The first measurement costs one extra run per baseline; the second costs a log line. Together they turn *our model scores higher* into *our model scores higher because it predicted contacts that actually happened*, which is the claim a physical-world result needs.

# 6 Limitations

The scenes are synthetic, planar, with three shapes and one object size, and no manipulator is involved; we have not tested the instrument on real video or on contact-rich interaction. Our perception reads a detector trained with the dataset's own labels. The simulators span a range of quality but all are rigid-body planar models, so we cannot say where saturation begins, only that it lies below the accuracy of straight-line motion. Seven checkpoints of one training recipe span 0.5 points on this benchmark, which is as large as several of the effects we report, and that spread is itself part of the finding. Whether the same insensitivity holds for benchmarks that grade embodied task success is the question we would most like to discuss.

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